2.5MHz 6V 5A Fast PWM Synchronous Step-Down Converter

DESCRIPTION

The MT3185 is a 5A high efficiency constant on-time controlled synchronous step-down converter. It operates with input voltage from 2.7V to 6V and provides output range from 0.8V to as high as input level, thanks to its 100% duty cycle operation. Its advanced constant on-time control Fast-PWM scheme simplifies loop compensation and offers excellent load transient response while maintaining a relatively constant 2.5MHz switching frequency. MT3185N consumes extremely low 17μA quiescent current hence achieves superior light load efficiency. The high gain error amplifier in the control loop provides excellent load and line regulation. For fault tolerant operation, MT3185 has cycle-by-cycle current limit protection and hiccup mode for short circuit or over-load condition.

MT3185 is available in 7 lead DFN2mmx2mm package and ideal for high performance, portable applications.

FEATURES

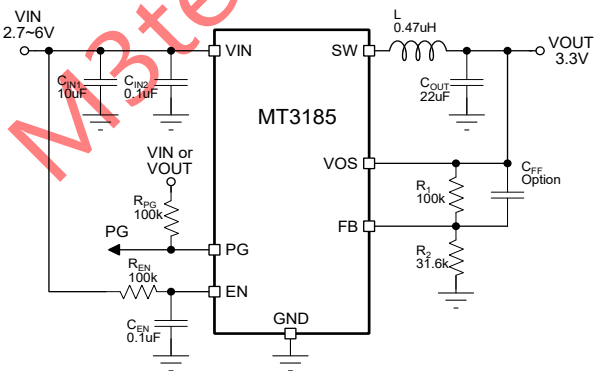
- Wide Input Range from 2.7V to 6V.
- MT3185N:2.5MHz with Pulse Skip Mode (PSM)
  MT3185A:2.5MHz with Forced PWM Mode (FPWM)
- 17μA Quiescent Current (MT3185N)
- Continuous 5A Output Current, Peak up to 6A.
- High Efficiency up to 97%
- Output Voltage as low as 0.8V
- 100% Duty Cycle Operation
- +/-1% 0.8V Feedback Voltage Accuracy
- 2.5MHz Pseudo Constant Switching Frequency
- Built-in 30mΩ HS and 25mΩ LS Power Switches
- Cycle-by-cycle Current Limit Protection
- Hiccup Mode for Short Circuit and Over-Load Protection
- Open Drain Power Good Indication with Internal Pull-up Resistor
- Output Discharging Function in Shutdown
- Thermal Shutdown Protection
- Stable with low ESR ceramic Output Capacitors
- Available in a Small DFN 2mmx2mm_7L Package
- Pb-Free RoHS Compliant

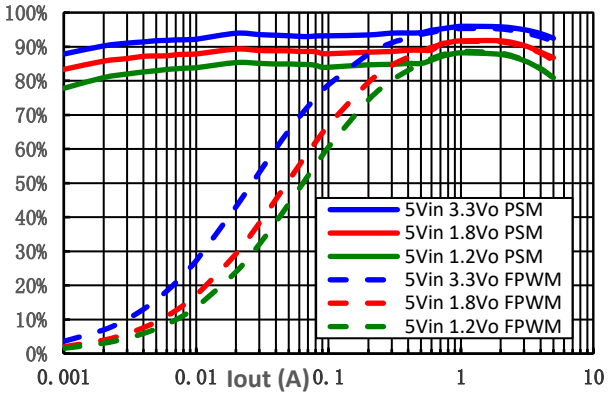
APPLICATIONS

- Solid-State and Hard Disk Drives
- Portable / Handheld Devices
- WiFi Module/ Set-top Boxes
- DC/DC Micro Modules

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TYPICAL APPLICATIONS



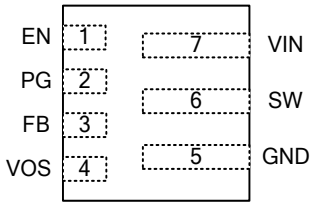


ORDERING INFORMATION

Part No.	Marking	Temp. Range	Remark	Package	MOQ
MT3185NFDR	3185 YWXX	-40°C ~+85°C	PFM Mode	DFN 2x2_7L	3000/Tape & Reel
MT3185AFDR	185A YWXX	-40°C ~+85°C	FPWM Mode	DFN 2x2_7L	3000/Tape & Reel

Note: Y: Year, W: Week, XX: Control Code

PIN CONFIGURATION



MT3185_DFN2X2_7LTOP VIEW

PIN DESCRIPTION

DFN2x2_7L PIN NO.	PIN NAME	DESCRIPTION
1	EN	Enable Control Input with accurate 1.21V enable threshold which can be used to build precision R-C turn-on delay and input under-voltage lockout. EN pin has an internal 500KΩ pull-down resistor to AGND. Drive EN above 1.21V to turn on the converter. Drive EN below 1.1V to turn off the converter and discharge output (Optional).
2	PG	Power Good Open-drain Output. Connect a 100kΩ pull-up resistor to V_{IN} or V_{OUT}.
3	FB	Feedback voltage input connect to external feedback resistors
4	VOS	Output voltage sense pin. This pin must be directly connected to the output capacitor.
5	GND	Analog ground and LS switch Power ground
6	SW	Switching pin, connect to the external inductor
7	VIN	Analog input supply voltage and HS Switch input Supply voltage

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Absolute Maximum Rating (Reference to GND) (Note1)

SW to GND ..... -0.5V to V_{IN}+1V
Dynamic V_{SW} in 10ns Duration ..... -2V to V_{IN} +2V
VIN to GND ..... -0.3V to +6.5V
EN, OUT, FB, PG to GND..... -0.3V to +6V
ESD ..... Class 2
Lead Temperature (Soldering 10s) ..... 260°C
Junction Temperature Range. .....-40°C to 150°C
Storage Temperature Range ..... -65°C to 150°C

Recommend Operating Conditions (Note2)

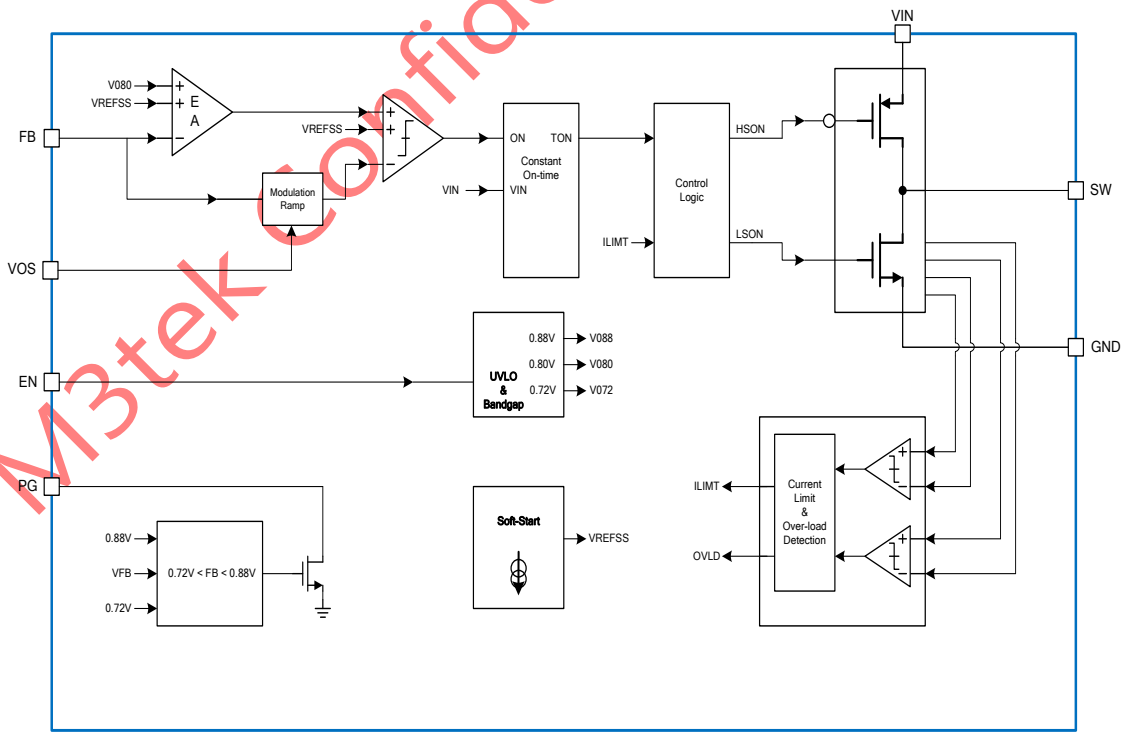
Input Voltage (V_{IN}) .....2.7V to 6V
Output Voltage (V_{OUT}) ..... +0.6V to V_{IN}
Operating Temperature Range .....-40°C to 85°C
Junction Temperature Range, T_J .....125°C

Thermal information (Note3, 4)

Maximum Power Dissipation (T_A=25°C) 1.79W
Thermal Resistance(θ_{JA}) .....69.7°C/W
Thermal Resistance(θ_{JC}) .....27.3°C/W

Note(1): Stress exceeding those listed "Absolute Maximum Ratings" may damage the device.
Note(2): The device is not guaranteed to function outside of the recommended operating conditions.
Note(3): Measured on JESD51-7, 4-Layer PCB.
Note(4): The maximum allowable power dissipation is a function of the maximum junction temperature T_{J_MAX}, the junction to ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by P_{D_MAX}= (T_{J_MAX}-T_A)/θ_{JA}. Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.

FUNCTIONAL BLOCK DIAGRAM



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ELECTRICAL CHARACTERISTICS

T_A = 25°C, V_{IN}=5V, unless otherwise noted. Typical values are at V_{BAT} = V_{EN} = 3.6V and V_{OUT} = 5V.

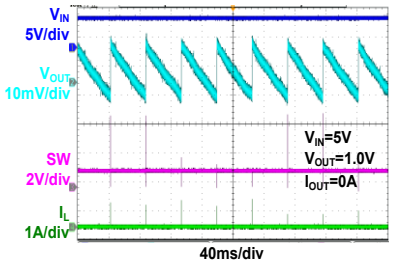
PARAMETER	SYMBOL	TEST CONDITIONS		MIN	TYP	MAX	UNIT
Input Voltage Range	V_{IN}			2.7		6	V
Shutdown Current	IISHD	V_{EN} = 0V, V_{IN} = 5.5V			0.1	5	μA
Input Under Voltage Lockout Threshold	V_{UVLO}	V_{IN} Increasing			2.4		V
Input Under Voltage Lockout Hysteresis	V_{UVHYS}				280		mV
Input Over Voltage Lockout Threshold	V_{INOV}	V_{IN} Increasing			6.5		V
Input Over Voltage Lockout Hysteresis	V_{INOVHY}				500		mV
Output Over-voltage Protection		V_{FB} with ramp to over voltage			+20		%
Quiescent Current	I_Q	V_{FB} = 0.83V	MT3185N		17		μA
			MT3185A		500		μA
Feedback Voltage	V_{FB}			792	800	808	mV
Feedback Current	I_{FB}				1		nA
HS Switch Current Limit	I_{LIM}				8		A
LS Switch Zero-Current Detection	I_{ZRO}	MT3185N Only			0.03		A
LS Switch Negative Current Limit	I_{NEG}	MT3185AOnly			-3		A
HS Switch On Resistance	R_{ONHS}				30		mΩ
LS Switch On Resistance	R_{ONLS}				25		mΩ
HS Leakage Current	I_{LKHS}	V_{IN} = 5.5V, V_{EN} = V_{SW}= 0V			0.1	10	μA
LS Leakage Current	I_{LKLS}	V_{IN} = V_{SW}= 5.5V, V_{EN} = 0V			0.1	10	μA
On-time for HS Switch		V_{IN} = 5V, V_{OUT} = 1.2V			95		ns
		V_{IN} = 5V, V_{OUT} = 3.3V			265		ns
Output Discharging Resistance	V_{PGLO}	V_{EN} = 0V, V_{OUT}= 1V			6		Ω
PGOOD Output Low Voltage		V_{FB} =0.5V, sink 1mA			0.2	0.3	V
PGOOD Output Leakage Current	I_{LKPG}	V_{FB} =0.63V, VPGOOD = V_{IN} = 5.5V			10	100	nA
PGOOD Under Voltage Rise Threshold	P_{CTPGR}	V_{FB} with ramp up from under voltage		-12	-10	-8	%
PGOOD Under Voltage Fall Threshold	P_{CTPGF}	V_{FB} ramp down from regulation			-15		%
PGOOD Delay	T_{DPGD}	PGOOD going High to Low			30		μs
EN On Threshold V_{EN_ON}				1.16	1.21	1.26	V
EN Off Threshold V_{EN_OFF}				1	1.10	1.2	V
EN Internal Pull-Down Resistor (Rpd)					500		kΩ
Soft Start Time					2.5		ms
Thermal Shutdown					160		℃
Thermal Shutdown Hysteresis					30		℃

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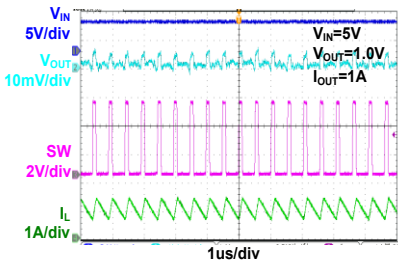
TYPICAL PERFORMANCE CHARACTERISTICS

C_{IN}=10uF+0.1uF,C=22uF, L=0.47uH, T_A=25℃

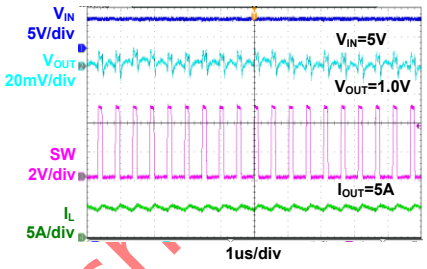
Steady State Test



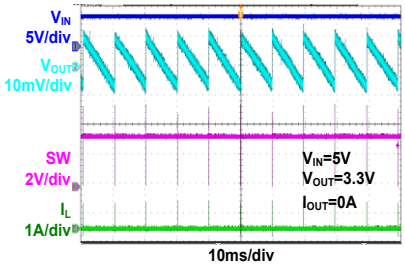
Steady State Test



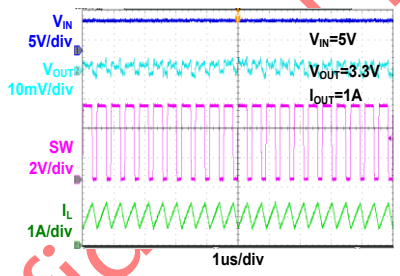
Steady State Test



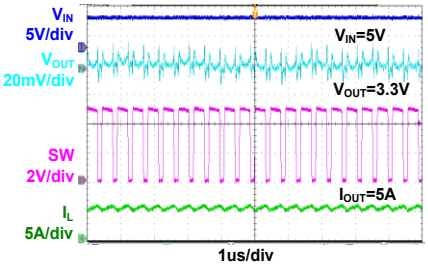
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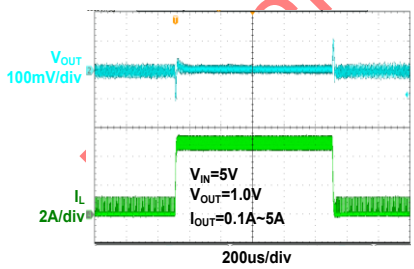
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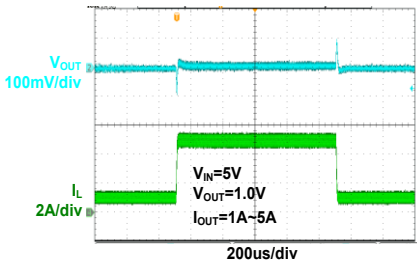
Steady State Test



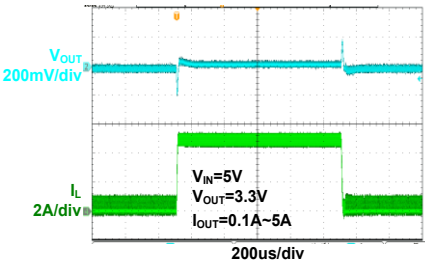
Load Transient Response



Load Transient Response



Load Transient Response

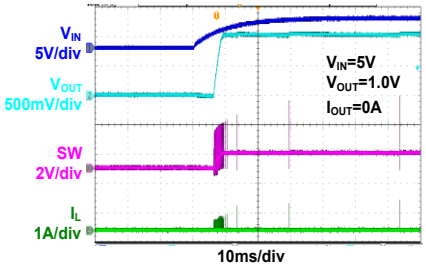


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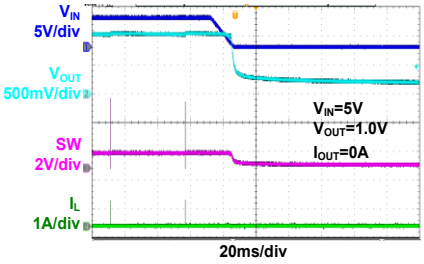
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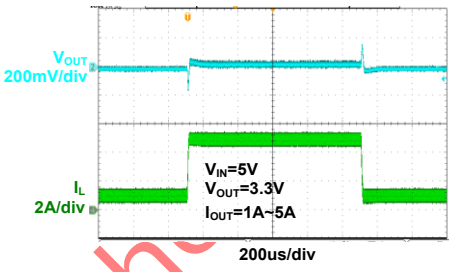
Vin Power On



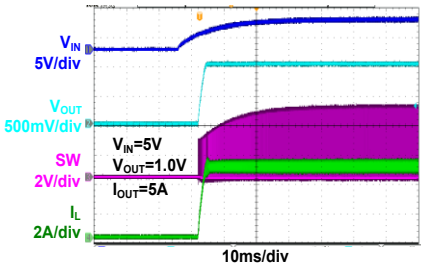
Vin Power Off



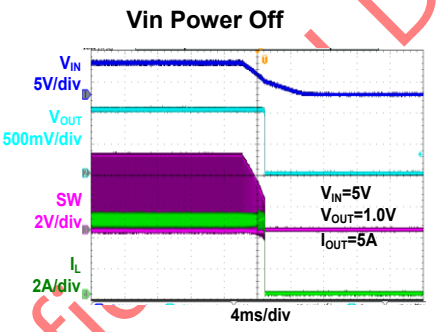
Load Transient Response



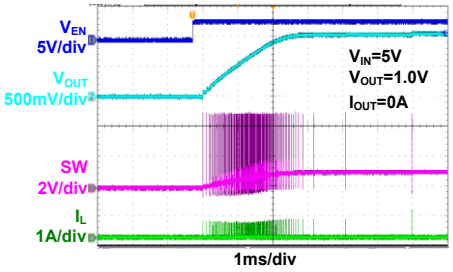
Vin Power On



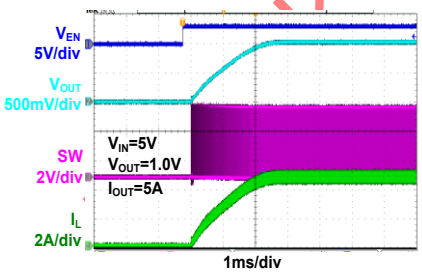
Vin Power Off



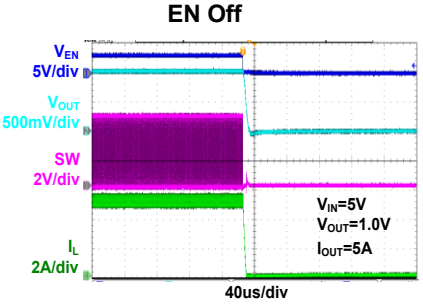
EN On



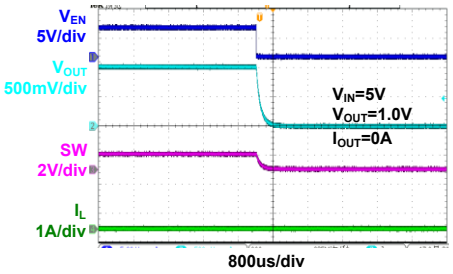
EN On



EN Off



EN Off

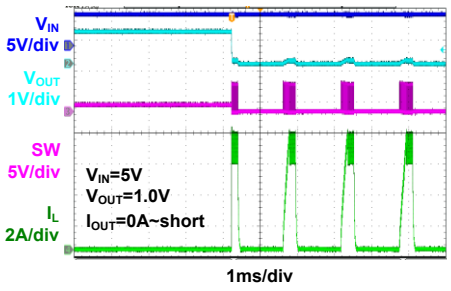


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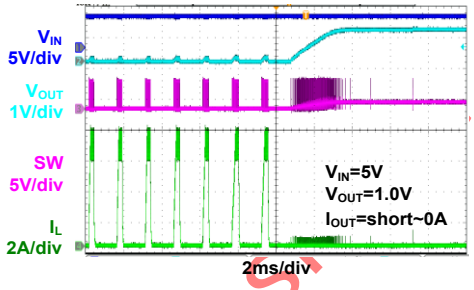
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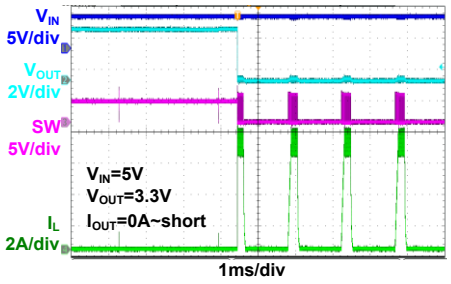
Short Circuit Protection



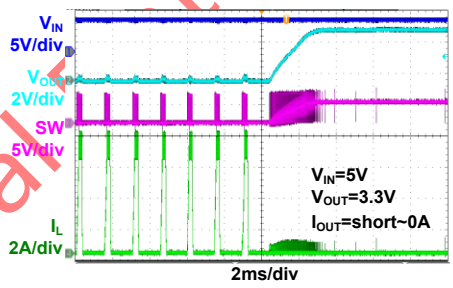
Short Circuit Recovery



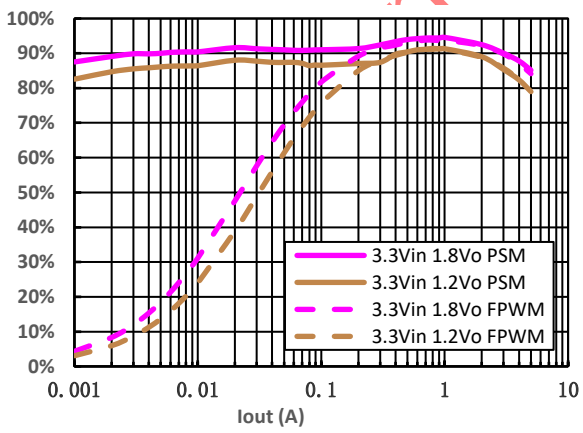
Short Circuit Protection



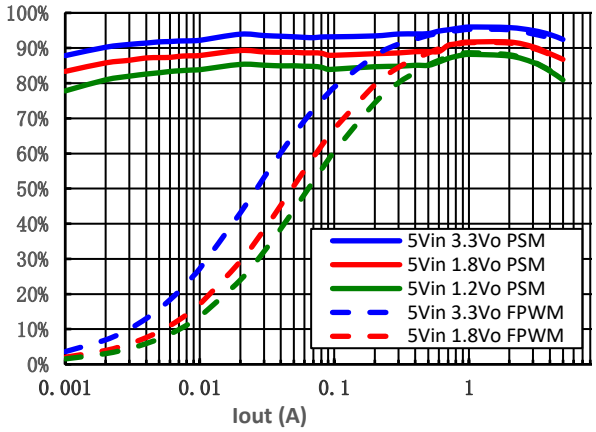
Short Circuit Recovery



Efficiency



Efficiency



Confidential - tas

OPERATION

MT3185 is a constant on-time controlled synchronous step-down converter that offers excellent transient response over a wide range of input voltage. It achieves superior light-load efficiency with extremely low quiescent current.

Fast -PWM Constant On-Time Control

MT3185 employs advanced constant on-time control (Fast-PWM) to achieve superior transient response. The Fast-PWM constant on-time control turns on HS immediately when FB drops below reference. The HS is turned on for a pre-determined period (on-time) to ramp up the inductor current, and then the LS will be turned on to ramp down the inductor current. The cycle repeats itself, if FB drops below reference again.MT3185 uses adaptive on-time based on input and output voltage level to achieve a relatively constant switching frequency. The on-time can be estimated as:

$$T_{ON} = \frac{V_{OUT}}{V_{IN}} \cdot 0.4\mu$$

Due to its immediate response on FB voltage droop and simplified loop compensation, Fast-PWM constant on-time offers a superior transient response compare to traditional fixed frequency PWM control step-down converters.

Light Load Operation

In light load condition where the switcher operates in discontinuous mode, MT3185 cuts down its quiescent current to as low as 17uA, thus achieve excellent light load efficiency.

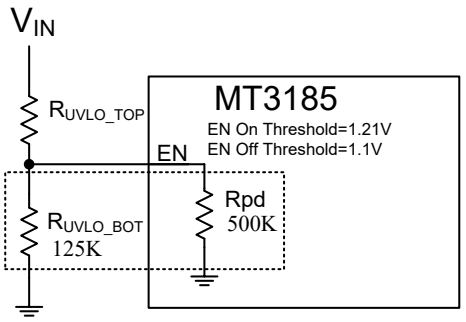
Enable

When input voltage is above the Under-Voltage Lock-out threshold, MT3185 can be enabled by pulling the EN pin to above 1.3V. MT3185 will be disabled if the EN pin is kept below 0.3V to get 0μA shutdown current. Enable Control Input pin EN with an accurate enable ON threshold V_{EN_ON} (Typical 1.21V) and enable OFF threshold VEN_OFF (Typical 1.10V) which can be used to build precision R-C turn-on delay and input under-voltage lockout. External adjustable input UVLO can be accomplished by using the EN pin. EN pin has an internal 500kΩ pull-down resistor (R_{PD}) to AGND. As shown in the below Figure, a resistor divider from the V_{IN} pin to GND sets the input UVLO level. Choose the bottom UVLO resistor R'_{UVLO_BOT} in the 10kΩ~200kΩ range to set the divider current at 10μA or higher. Typically select R'_{UVLO_BOT}=100kΩ. The value of top resistor R_{UVLO_TOP}, depending on the desired turn-on voltage V_{UVLO_Rising} (or $V_{UVLO_Falling}$) at the V_{IN} pin, can be calculated with below Equation.

$$V_{UVLO_{Rising}} = V_{EN_ON} \times \left(\frac{R_{UVLO_TOP} + R'_{UVLO_BOT}}{R'_{UVLO_BOT}} \right)$$

$$R'_{UVLO_BOT} = R_{UVLO_BOT} // R_{PD}$$

$$V_{UVLO_{Falling}} = V_{EN_OFF} \times \left(\frac{R_{UVLO_TOP} + R'_{UVLO_BOT}}{R'_{UVLO_BOT}} \right)$$



R' UVLO_BOT=RUVLO_BOT //Rpd
=125K//500K=100K

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Internal Soft Start

MT3185 has built-in 2.5ms soft-start timer. During the soft start period, output voltage is ramped up linearly to the regulation voltage, independent of the load current level and output capacitor value.

Current Limit and Hiccup Mode

MT3185 has cycle-by-cycle HS current limit protection to prevent inductor current from running away. Once HS current limit is triggered, MT3185 will turn on LS and wait for the inductor to drop down to a pre-determined level before the HS can be turned on again. If this current limit condition is repeated for a sustained long period of time, MT3185 will consider it as over-load or short circuit. Either way, MT3185 will enter hiccup mode, where it stops switching for a pre-determined period of time before automatically re-try to start up again. It always starts up with soft-start to limit inrush current and avoid output overshoot.

Power Good Indication

MT3185 has open drain PG indicator with internal 1MΩ pull-up resistor to input voltage. PG will be pulled up if output voltage is within +/-10% of regulation, otherwise PG is pulled down by the internal NMOS.

APPLICATION INFORMATION

Setting the Output Voltage

External feedback resistors are used to set the output voltage. Refer to typical application circuit on page 1, the bottom feedback resistor R_2 has some impact on the loop stability, so its recommended range is below 100kΩ. For any chosen R_2, the top feedback resistor R_1 can be calculated as:

$$V_{OUT} = V_{REF} \cdot \left(\frac{R_1}{R_2} + 1\right), \text{thus } R_1 = R_2 \cdot \left(\frac{V_{OUT}}{0.8} - 1\right)$$

Inductor Selection

The recommended inductor value for MT3185 is between 0.47uH to 4.7uH. Usually the inductor value is chosen to satisfy a desired ripple current:

$$L = \frac{V_{OUT} \cdot (V_{IN} - V_{OUT})}{V_{IN} \cdot f_{SW} \cdot \Delta I}$$

Where ΔI is the inductor ripple current.

With the chosen ΔI, the peak inductor current will be:

$$I_{PK} = I_{LOAD} + \frac{1}{2} \cdot \Delta I$$

Input Bypass Capacitor Selection

The input current to the step-down converter is discontinuous with very sharp edges, therefore an input bypass capacitor is required. For best performance, it's recommended to use low ESR ceramic capacitors and place them as close to the input pin as possible. For lowest temperature variations, use X5R or X7R dielectric ceramic capacitors. The RMS current of the input capacitor is approximately:

$$I_{CIN_{RMS}} = I_{OUT}\sqrt{D(1-D)}$$

From the equation, it can be seen that the highest RMS current occurs when D is 0.5:

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$$I_{CIN_{RMS}} = \frac{1}{2} I_{OUT}$$

Choose the capacitor with RMS current rating higher than 1/2 I_{OUT}. The power dissipation on the input capacitor can be estimated with the RMS current and the ESR resistor.

Electrolytic or tantalum capacitors can also be used, but due to their significantly higher ESR, a small size ceramic capacitor should be placed as close to the IC as possible.

The voltage ripple on the input capacitor, neglecting the ESR impact, can be calculated as:

$$\Delta V_{CIN} = \frac{I_{LOAD}}{f_{SW} \cdot C_{IN}} \cdot \frac{V_{OUT}}{V_{IN}} \cdot \left(1 - \frac{V_{OUT}}{V_{IN}}\right)$$

Output Capacitor Selection

An output capacitor is required to obtain a stable output voltage. To minimize the output voltage ripple, ceramic capacitors should be used, and the ripple voltage can be estimated as:

$$\Delta V_{OUT} = \frac{1}{8} \cdot \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \cdot \frac{V_{OUT}}{L} \cdot \frac{1}{(f_{SW})^2 \cdot C_{OUT}}$$

If electrolytic or tantalum capacitors are used, the ESR will dominate the output voltage ripple:

$$\Delta V_{OUT} = \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \cdot \frac{V_{OUT}}{f_{SW} \cdot L} \cdot R_{ESR}$$

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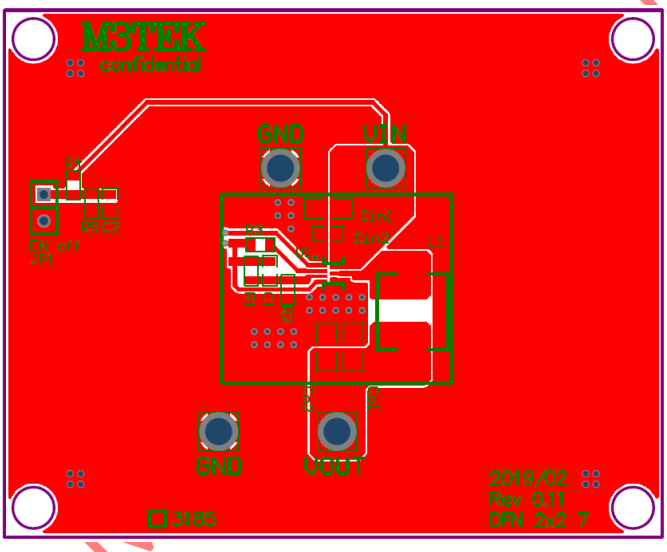
PCB Layout Recommendation

Proper layout and component placement are very important, and sometimes critical for the step-down converters. The PCB can radiate excessive noise and contribute to converter instability with improper layout.

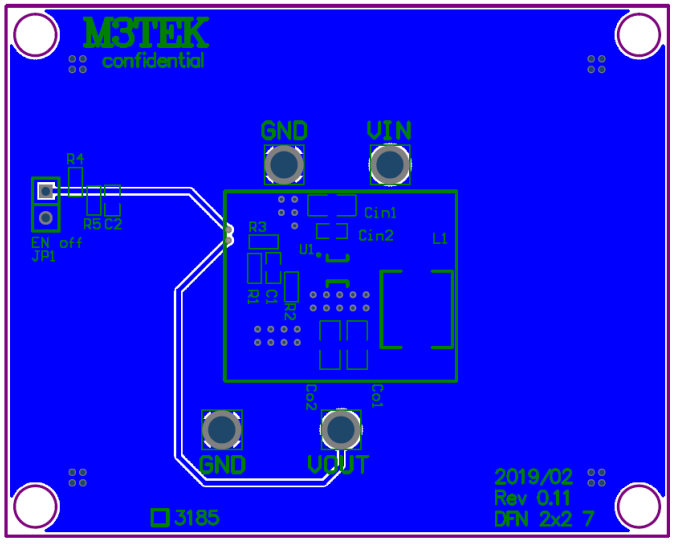
Certain points must be considered before starting a layout using the MT3185.

- Use wide trace for the high current paths, the traces of the high current paths as short and wide as possible.
- Place the input capacitor C_{IN} close to PVIN and PGND, so the total PCM trace from the pins to C_{IN} capacitor is minimized.
- SW pin and the trace will be the noisiest signal on the board, so proper isolation between SW from FB is essential, SW node encounters high frequency voltage swings, so it should be kept in a small area.
- Ensure all feedback network connections are short and direct. Place the feedback network as close to the chip as possible.
- Components away from the SW node to prevent stray capacitive noise pick-up.
- The GND pin and Exposed Pad should be connected to a strong ground plane for heat sinking and noise protection.
- An example of PCB layout guide is shown in below Figure for reference.

Preliminary Datasheet

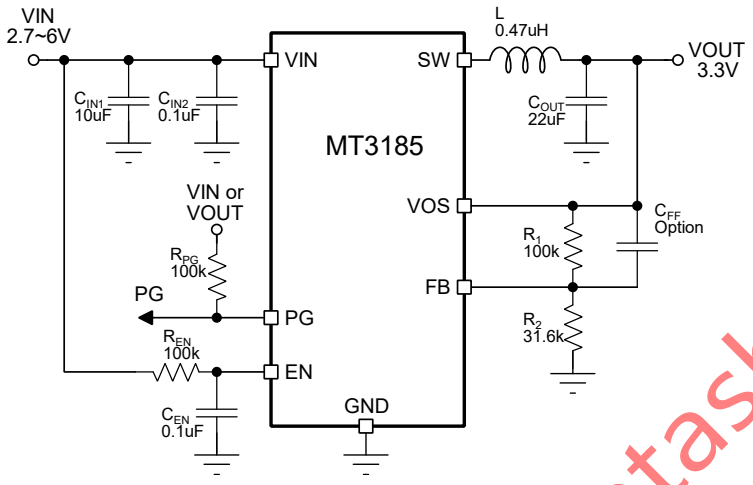


Top Layer



Bottom Layer

Application Schematic



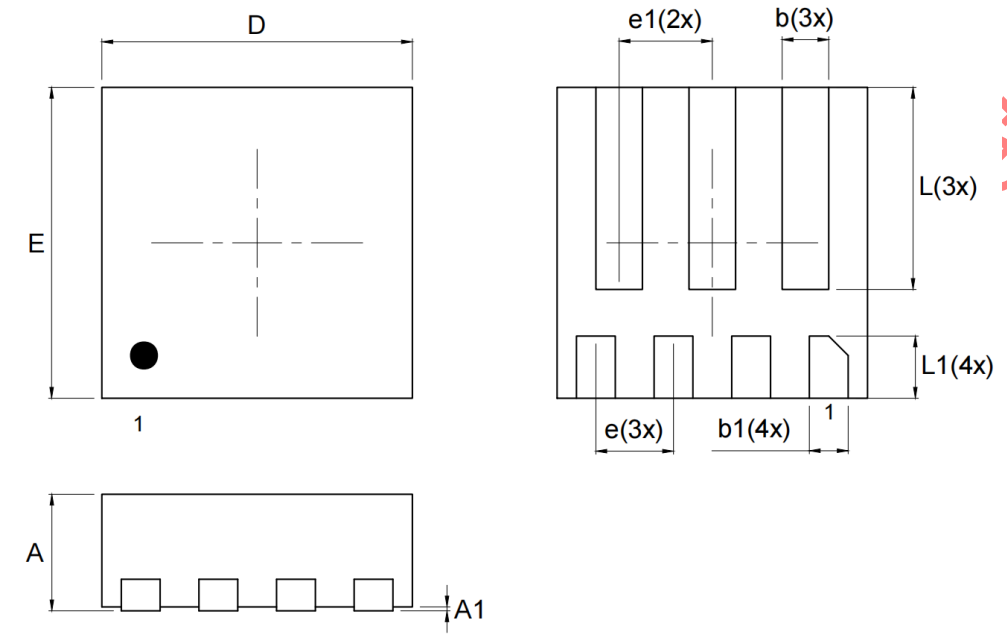
EVB BOM List

Qty	Ref	Value		Description	Package
1	C_{IN1}	10μF		Ceramic Capacitor, 10V, X5R	0805
1	C_{IN2}	0.1μF		Ceramic Capacitor, 10V, X5R	0603
1	Cout	22μF		Ceramic Capacitor, 10V, X5R	0805
1	Rpg	100kΩ		Resistor, ±1%	0603
1	R_{EN}	100kΩ		Resistor, ±1%	0603
1	C_{EN}	0.1μF		Ceramic Capacitor, 10V, X5R	0603
1	L	0.47uH		Inductor:MPCA-0630-R47-M, 4.13mohm	SMD
1	R1	Vout=3.3V	100KΩ	Resistor, ±1%	0603
		Vout=1.8V	37.4KΩ		
		Vout=1.2V	15KΩ		
		Vout=1.0V	7.5KΩ		
1	R2	Vout=3.3V	31.6KΩ	Resistor, ±1%	0603
		Vout=1.8V	30KΩ	Resistor, ±1%	0603
		Vout=1.2V	30KΩ	Resistor, ±1%	0603
		Vout=1.0V	30KΩ	Resistor, ±1%	0603
1	U1	MT3185		Step-Down DC/DC Converter	DFN2x2_7L

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PACKAGING INFORMATION

DFN 7L 2mmx2mm PACKAGE OUTLINE DIMENSIONS

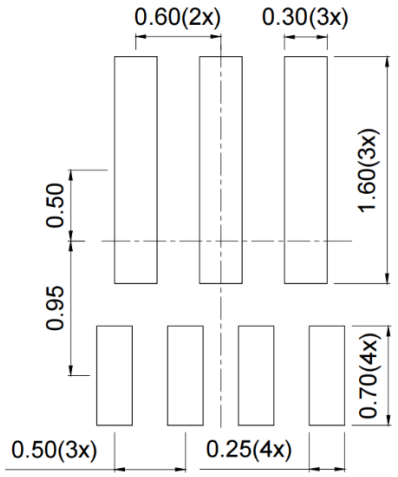


SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	0.70	0.80	0.028	0.031
A1	0.00	0.05	0.000	0.002
b	0.25	0.35	0.010	0.014
b1	0.20	0.30	0.008	0.012
E	1.90	2.10	0.075	0.083
D	1.90	2.10	0.075	0.083
L	1.25	1.35	0.049	0.053
L1	0.35	0.45	0.014	0.018
e	0.50 BSC		0.020 BSC	
e1	0.60 BSC		0.024 BSC	

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LAND PATTERN INFORMATION

DFN2x2_7L



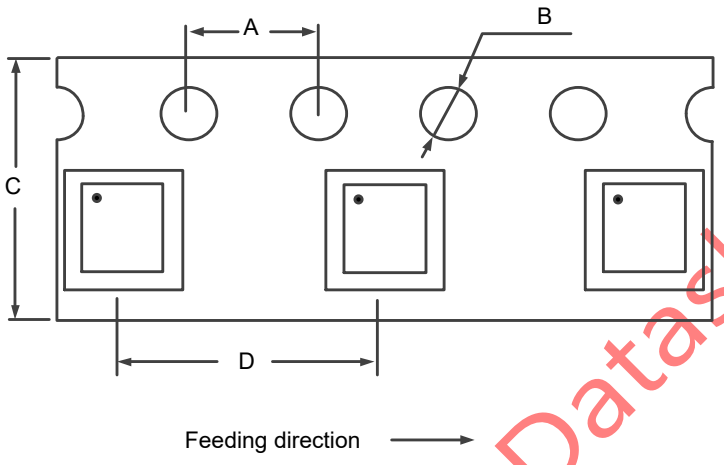
Note:

- Dimensions in mm.
- For reference only.

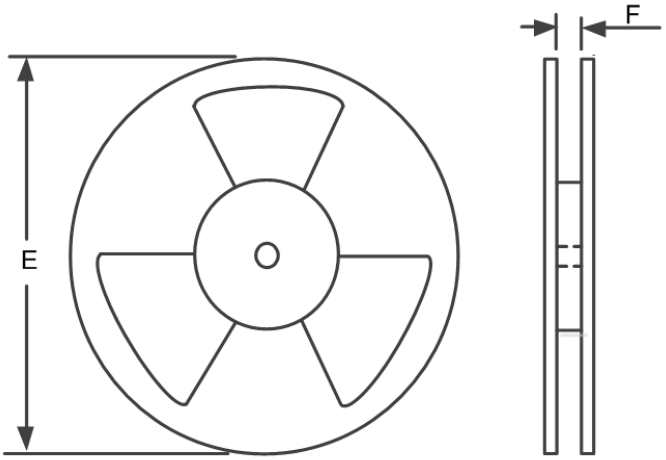
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CARRIER TAPE & REEL DIMENSIONS

1. Orientation / Carrier Tape Information:



2. Reel Information:



3. Dimension Details:

PKG Type	A	B	C	D	E	F	Q'ty/Reel
DFN 2x2	4.0mm	1.5mm	8.0mm	4.0mm	7inches	9.0mm	3,000

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REFLOW PROFILE

Classification of IR Reflow Profile

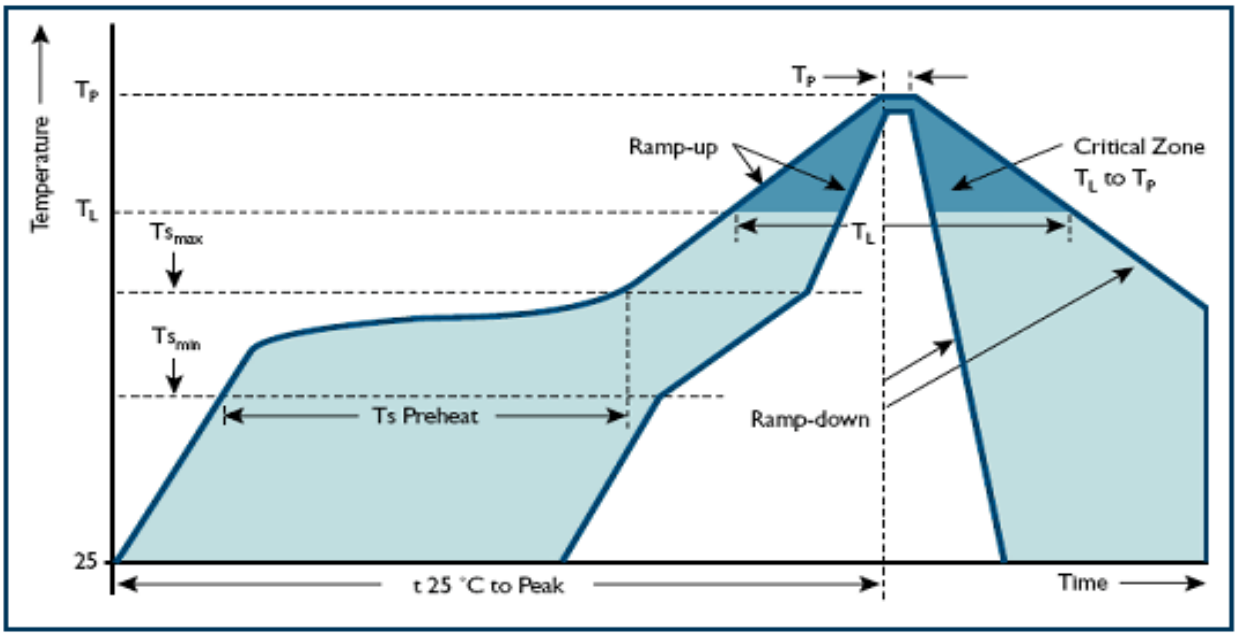
Reflow Profile	Green Assembly
Average Ramp-Up Rate (Ta_{smin} to Tp)	1~2℃/second, 3℃/second max.
Preheat & Soak -Temperature Min(Ts_{min}) -Temperature Max(Ts_{max}) -Time($ts_{min}$to $tsts_{max}$)	 150℃ 200℃ 60~120 seconds
Time maintained above: -Temperature(T_L) -Time(t_L)	 217℃ 60~150 seconds
Peak Temperature(Tp)	See Classification Temp intable1
Time within 5℃of actual Peak Temperature(tp)	30 seconds max.
Ramp-Down Rate	6℃/second max.
Time 25℃ to Peak Temperature	8 minutes max.

* Tolerance for peak profile Temperature (T_p) is defined as a supplier minimum and a user maximum.
** Tolerance for time at peak profile temperature (t_p)is defined as a supplier minimum and a user maximum.

Table 1. Pb-free Process –Classification Temperatures (Tc)

Package Thickness	Volume mm^3 <350	Volume mm^3 350-2000	Volume mm^3 >2000
<1.6mm	260℃	260℃	260℃
1.6mm–2.5mm	260℃	250℃	245℃
≥2.5mm	250 ℃	245℃	245℃

Note: For all temperature information, please refer to topside of the package, measured on the package body surface.



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